MIC501-Microbiology

Practical Exam Notes

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Subjective Questions

1. **What are the different biosafety levels and what precautions are taken at each level?**

Biosafety levels 1-4 have increasing levels of precautions. BSL-1 involves basic training and equipment. BSL-2 adds restrictions on access and uses safety cabinets. BSL-3 involves strict training and containment procedures. BSL-4 has the strictest containment methods for highly dangerous pathogens.

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2. **Name three pieces of equipment used in a microbiology lab and their purpose.**

Equipment used includes inoculating loops to transfer microorganisms, a microscope to view microorganisms, and an autoclave to sterilize media and equipment using steam heat and pressure.

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3. **What is the purpose of Gram staining and how does it work?**

Gram staining differentiates between Gram positive and Gram negative bacteria. It uses a primary stain, mordant, decolorizer, and secondary stain. Gram positive bacteria retain the primary stain while Gram negative bacteria take up the secondary stain.

4. **What safety precautions should be taken when performing antimicrobial sensitivity testing?**

Use aseptic technique, flame sterilize inoculating tools between use, avoid cross-contamination between samples, use protective equipment like gloves and goggles.

5. **What are four types of culture media used in microbiology?**

Types include synthetic (chemically defined), non-synthetic (not fully defined), selective (inhibits certain bacteria), differential (allows distinction of bacteria by colony characteristics).

6. What are four phases of the bacterial growth curve?

The four phases are lag phase with no growth, log/exponential rapid growth phase, stationary phase where growth plateaus, and death phase where cell numbers decline.

7. How can microbial cell numbers be quantified by plating?

Cells are spread on agar plates to grow into colonies arising from individual cells that can be counted. Cell concentration is calculated using the colony count and the dilution factor.

8. What is the purpose of biochemical tests in microbiology?

Biochemical tests analyze bacteria's ability to utilize substrates, produce metabolites and enzymes. They help differentiate between bacterial groups and aid identification.

9. What precautions should be taken when sterilizing by autoclaving?

Place caps loosely to allow steam penetration and prevent explosion, use recommended temperatures and times (121°C for 15 min), allow cooling before opening to prevent burns.

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10. How does the Gram staining procedure differentiate bacteria?

Gram positive bacteria retain the crystal violet dye after decolorization while Gram negative bacteria lose the dye and take up the safranin counterstain, appearing red under the microscope.

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11. What are four physical methods of sterilization?

Heat methods include autoclaving with steam, dry heat ovens, and incineration. Radiation methods include UV light and ionizing radiation. Filtration also physically removes microbes.

12. What is the purpose of microscopy in studying microorganisms?

Microscopy allows visual examination of microbial cell, colony, and growth characteristics to aid identification and understanding of microbes. Different microscopy methods provide enhanced details.

13. Name the four IMViC biochemical tests. What does each test for?

The IMViC tests are: Indole (tryptophanase enzyme), Methyl Red (mixed acid fermentation), Voges-Proskauer (butanediol fermentation), and Citrate (citrate utilization).

14. What safety guidelines should be followed when working at Biosafety Level 2?

Guidelines include restricted lab access, biohazard signs, impervious lab surfaces, presence of a safety cabinet and autoclave, wearing gloves and lab coat.

15. What are selective and differential culture media used for?

Selective media inhibits unwanted bacteria, enrichment media facilitates pathogen growth. Differential media allows distinction of bacteria based on colony characteristics.

16. What are four chemical sterilization methods?

Chemical methods include alcohols like ethanol, aldehydes like formaldehyde, oxidizing agents like hydrogen peroxide, and ethylene oxide gas.

17. What are the components of a light microscope?

Components include ocular lens, objective lenses, stage, mirror/light source, condenser, diaphragm to adjust light intensity, coarse and fine adjustment knobs.

18. What safety guidelines should be followed when performing microbiology experiments?

Guidelines include following aseptic technique, avoiding contamination between samples, flaming inoculating tools between use, wearing protective equipment, and properly disposing waste.

19. What is the purpose of the Kirby-Bauer antibiotic sensitivity test?

It determines bacterial sensitivity or resistance to various antibiotics, seen as zones of inhibition on agar plates. This guides effective antibiotic treatment.

20. What are mechanical or filtration methods for microbial sterilization?

Filtration uses membrane filters or sintered glass with specific pore sizes to physically remove bacteria from thermolabile solutions and gases.

21. What are the characteristics examined when studying colony morphology of bacteria?

Colony characteristics examined include shape, elevation, margin, size, surface appearance, texture/consistency, pigmentation, and opacity.

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22. What is the purpose of a Durham tube in carbohydrate fermentation testing?

A Durham tube collects gas produced from carbohydrate fermentation, allowing differentiation between fermentative and non-fermentative organisms.

23. How can citrate utilization be detected through biochemical testing?

Growth on Simmons citrate agar turns the bromothymol blue indicator from green to blue, signalling alkaline byproducts from citrate breakdown.

24. What is the principle behind using turbidity measurement to quantify microbial growth?

Turbidity is proportional to cell density - less light penetrates a more turbid, microbial growth-rich culture broth. An optical density growth curve relates turbidity to cell numbers.

25. What are four moist heat sterilization methods?

Moist heat methods include boiling, autoclaving, tyndallization (sequential boiling), and pasteurization used for milk. The steam penetrates cells to kill vegetative cells and spores.

26. What is the purpose of a spreader when plating out cells on nutrient agar?

The spreader evenly distributes cells over the plate to prevent overlapping colonies and obtain isolated colonies each arising from an individual cell.

27. What are four dry heat sterilization techniques?

Dry heat techniques include incineration, dry heat ovens, flaming inoculation loops, and glass bead sterilizers relying on high temperatures rather than steam.

28. What is the purpose of a control when performing antimicrobial sensitivity testing?

The control without antibiotic evaluates normal growth to compare with any growth inhibition caused in samples with antibiotics, confirming antibiotic effects.

29. What biological risks are assessed when assigning biosafety levels?

Risks considered include severity and transmission potential of disease from the organisms, availability of preventives and treatment. Stringent containment increases with higher risks.

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30. What is the purpose of selective and differential culture media in the microbiology laboratory?

They select for growth of desired organisms while inhibiting contaminants and allow differentiation of mixed cultures by colony characteristics.

31. What safety precautions should be taken when using an autoclave?

Wear insulated gloves when handling hot materials, allow pressure to stabilize before opening, keep faces away when opening, and allow liquids to stop boiling before removing.

32. What are chemically defined and non-defined culture media and their differences?

Chemically defined media contain known purified ingredients while non-defined contain complex ingredients like peptones with variable composition.

33. What are the main outcomes of carbohydrate fermentation testing?

Outcomes are acid production signaling carbohydrate metabolism, visualized through a pH indicator color change, and gas production from more extensive fermentation.

34. What are some oxidase test positive organisms and some oxidase test negative organisms?

Positive: Pseudomonas, Aeromonas, Campylobacter, Pasteurella. Negative: Enterobacteriaceae including E. coli and Klebsiella.

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35. Which sterilization methods can destroy bacterial endospores and which cannot?

Moist heat (autoclaving) and ionizing radiation effectively kill endospores. Dry heat and alcohol do not reliably kill endospores.

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36. What are biosafety cabinets and their purpose in containment of biological hazards?

Biosafety cabinets provide enclosed sterile workspaces, protecting the experimenter and preventing release of pathogens into the wider laboratory environment.

37. What phenotypic characteristics are revealed through IMViC testing?

Butanediol and mixed acid fermentation pathways for glucose metabolism, citrate utilization as sole carbon source, and ability to produce indole from tryptophan breakdown.

38. What is the purpose of a Kirby-Bauer antibiotic susceptibility test?

It determines antibiotic efficacy against a bacterial isolate based on zones of inhibition of growth around antibiotic-impregnated discs on agar.

39. What are some applications of membrane filtration microbial sterilization?

Heat-labile solutions like serum, antibiotics, vitamins and vaccines can be sterilized by filtration without degradation by heat or chemicals.

40. What is the purpose of a spreading tool used in microbiological spread plating techniques?

It evenly distributes cells over the agar surface to obtain isolated colonies that arise from individual cells after incubation.

41. Describe biosafety levels 1-4, including the organisms handled at each level and containment requirements.

Biosafety level 1 works with non-pathogenic organisms and require basic precautions like no eating/drinking, handwashing, etc. BSL2 handles moderate risk pathogens transmitted through ingestion/punctures and adds restricted access and biohazard signs. BSL3 has serious disease-causing pathogens requiring full protective gear, air filtration, training in handling infectious materials, and more stringent waste decontamination steps. BSL4 houses extremely dangerous easily-transmissible pathogens causing fatal untreatable disease, requiring maximum containment precautions like airlocks, vacuum lines, isolated building sites, full-body positive air pressure suits, and rigorous disinfection.

42. Discuss various culture media used in microbiology, their composition, types, uses, and some examples.

Culture media provides nutrients and physical growth conditions for microbial growth. They can be chemically defined with known ingredients or non-defined with complex ingredients like peptones. Types include basal media supporting general growth, enriched providing extra nutrients, selective allowing growth of select microbes, indicators differentiating bacteria by colony characteristics, and differential distinguishing bacteria biochemically. E.g. Nutrient media for general growth, MacConkey with bile salts selecting intestinal microbes and inhibiting others, mannitol salt agar for staphylococci, blood agar for fastidious organisms and hemolytic patterns. They are available in liquid broths aiding larger scale growth or solid agar media allowing isolated colonies.

43. Describe sterilization methods in detail including moist heat, dry heat, radiation, and filtration approaches. Cover the working principle, procedures used, and advantages/disadvantages of each.

Moist heat methods like autoclaving and tyndallization use steam penetration under pressure to kill vegetative cells and spores. Dry heat methods like incineration, baking, flaming, glass bead sterilizers use high temperatures to destroy vegetative cells but not necessarily spores. Radiation methods involve exposure to ionizing radiation like gamma-rays and non-ionizing U.V, effectively sterilizing material. Filtration uses membrane filters or sintered glass allowing thermolabile solutions to pass through while physically removing microbial contaminants. Factors favouring a method are feasibility of heat/radiation resistant samples, required turnaround times, capital costs and running expenses. Sterilization needs balancing with retaining sample integrity and biological activity.

44. Explain the Kirby-Bauer antibiotic susceptibility test performed in microbiology. Include the objectives, working principle, procedure, interpretation of results and applications.

The Kirby-bauer test evaluates antibiotic efficacy against a bacterial isolate to guide optimal treatment. Standardized antibiotic discs are placed on agar plates swabbed with a culture and incubated. Zones of inhibition around each disc result from diffusion of antibiotics and inhibition of bacterial growth. Zone diameters correlate to clinical breakpoints categorizing the bacteria as susceptible, intermediate or resistant to the antibiotic concentration in the disc. Comparing zones of multiple antibiotics helps determine the most appropriate single or combination antibiotic therapy for a bacterial infection in a patient.

45. Discuss various biochemical tests performed in microbiology with their objectives, principles, procedures, and interpretation of results.

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Biochemical tests assess microbial metabolic capabilities, supplementing morphological, culture and growth characterizations. E.g. carbohydrate fermentation detects acid and gas production signalling utilization pathways, IMViC tests check glucose metabolism via mixed acid/butanediol pathways, indole and citrate utilization, catalase and oxidase tests detect enzyme activities. Procedures involve inoculating media containing substrates for the specifically tested biochemical pathway and indicators to allow visualization of products generated. Results are interpreted as positive or negative for the ability to metabolize substrates through certain pathways. Biochemical profiling aids identifying unknown organisms and understanding their physiology alongside genetic studies.

46. Describe the preparation and outcomes of a bacterial growth curve over time performed in the microbiology laboratory. What information does it provide about bacterial growth phases?

Bacterial growth curves plot cell number or culture turbidity measurements over time. Lag phase shows no growth as cells adapt to the new conditions. Log/exponential phase displays rapid doubling of turbidity/cell count as binary fission proceeds unchecked. Stationary phase reaches a plateau with maximum cell density where growth equals death. Death/decline phase occurs as nutrients diminish, waste accumulates and viability reduces. The curve allows calculation of generation time, comparing growth rates across conditions, identifying ideal harvest points, and recognizing contamination. Growth phase studies facilitate optimizing production of biotechnology products like enzymes or antibiotics.

47. Discuss various methods of microbial cell count quantification performed in the microbiology laboratory with their working principles, procedures, and applications

Spread plate technique dilutes cultures across agar plates allowing development of isolated colonies that arise from individual cells that can be counted. Counts determine cells/ml through colony forming units and dilution factors. Most probable number estimates viable bacterial density through probability calculations of positive growth responses in replicate culture tubes after inoculation with dilutions giving terminal responses. Optical density measures culture turbidity spectrophotometrically, proportional to cell density as established through simultaneous cell counting. Automated systems detect impedance changes as cells pass electrodes placed amongst dilutions. Flow cytometry optically detects individual cell characteristics in a rapid fluid stream. All methods facilitate experiments tracking microbial population dynamics.

48. Describe the Gram staining procedure performed in microbiology along with its principle and interpretation. What is the significance of this differential staining technique?

Gram staining distinguishes bacterial cell wall structure. Crystal violet dye/iodine complexes penetrate all cells. Decolorizer removes dye from thin Gram negative walls while thick Gram positive walls retain it. Secondary safranin counterstain colours decolorized Gram negatives red. Gram positives hence appear purple while negatives are red under the microscope. The test aids rapid preliminary classification and identification of bacteria like streptococci and enterobacteria. Shape and clustering patterns provide further information alongside biochemical profiling. As cell wall targets, the response predicts susceptibility to certain antibiotics.

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49. What safety precautions need to be followed when working in a microbiology laboratory? Describe biosafety levels, equipment like biosafety cabinets, and practices upholding aseptic conditions.

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Risks from infective microbes require extensive safeguards and training. Practices include restricting lab access, hazard signs and equipment, monitoring personnel health, following standard operating procedures for equipment usage, techniques minimizing aerosols/splashes and surface decontamination. Safety equipment includes biosafety cabinets ensuring aseptic conditions, pipetting devices preventing mouth contact, biohazard containers for waste disposal after autoclaving/incineration. Further precautions adhere to ascending biosafety levels 1-4 depending on organism risk profiles. Safety practices aim to protect lab workers, wider environments and maintain experimental integrity against contamination.

50. What are the different methods of microbial sterilization and disinfection used in microbiological practice? Compare their mechanisms of action over vegetative cells, endospores, effectiveness and applications.

Physical methods like moist heat disrupts cells through protein coagulation and membrane damage while dry heat denatures them at longer exposures. Oxidizing chemical disinfectants attack cell membranes and enzymes. Heavy

metals and phenolic compounds disrupt membranes, enzymes and metabolism. Cationic detergents solubilize lipids and proteins in cell membranes. Alcohol rapidly penetrates to denature proteins. Combined incremental effects achieve disinfection at lower concentrations than sterilization. Sterilization necessitates higher concentrations and longer contact periods to destroy resistant endospores through multilayered action over germination, cortex and core regions. Method choice balances effectiveness against sample integrity preservation.

Multiple Choice Questions

1. What specifies the biosafety levels standards for handling infectious agents in the laboratory? b) CDC
2. Which biosafety level requires personnel to wear a one-piece positive pressure suit? d) BSL-4
3. What is the minimum holding time for moist heat sterilization by autoclaving at 121°C? c) 15 minutes
4. What is the temperature maintained in a hot air oven for sterilization of glassware? d) 160°C
5. Which of the following can NOT be sterilized by membrane filtration? b) Alcohols
6. Enriched media contains supplements like _____ to grow nutritionally demanding bacteria. d) All of the above
7. MacConkey agar is an example of _____ media.

c) Differential

8. Solid media contains agar at a concentration of _____. c) 1.5-2%
9. Broth medium is _____ in consistency.

c) Liquid

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10. The dye used in Gram staining to stain Gram positive bacteria is: b) Crystal violet

11. In the IMViC tests, "M" stands for:

a) Methyl red

12. A positive result in the Voges-Proskauer test is indicated by _____ coloration. c) Brownish-red

13. Citrate utilization test determines bacteria's ability to use _____ as sole carbon source. c) Sodium citrate

14. Which of the following biochemical tests detects presence of the enzyme catalase? c) Catalase

15. _____ is an electron transport chain enzyme detected in Oxidase test. b) Cytochrome c oxidase

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16. The substrate used in the urease test is:

b) Urea

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17. In bacterial growth curve, log phase is also called _____ phase. c) Exponential

18. Generation time measures the time required to achieve _____ of bacterial population. c) Doubling

19. Turbidity measurement is used to estimate _____ in bacterial cultures. b) Cell numbers

20. Primary stains are _____ stains using dyes like crystal violet. d) Simple

21. Hot air oven works at temperatures around _____ for sterilization.

d) 160°C

22. MacConkey agar differentiates bacteria based on _____ fermentation. b) Lactose

23. In citrate test, a positive result is indicated by _____ color change.

b) Green to blue

24. Positive result in methyl red test is shown by _____ coloration. b) Red

25. Indole test detects bacterial ability to split _____ amino acid.

b) Tryptophan

26. Motile bacteria generally form colonies with _____ morphology. b) Irregular

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27. An example of differential stain is _____.

d) Gram stain

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28. Anaerobic cultivation employs culture systems like _____.

b) Agar stabs

29. Growth of bacteria occurs in colonies on _____ solid media. c) Agar

30. Colonial margin of bacteria helps determine _____.

b) Motility

31. Positive result in oxidase test is indicated by _____ coloration. b) Purple

32. VP test detects production of _____, a precursor of 2,3-butanediol. c) Acetoin

33. Liquid or broth media is used for _____.

c) Mass cultivation of cells

34. Bacterial smears are heat fixed prior to Gram staining to _____.

b) Improve adherence

35. Selective media contains agents like _____ to inhibit unwanted bacteria. b) Antibiotics

36. Physical methods of sterilization include _____.

d) All of the above

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37. SIM media is used for _____ testing of microbes.

a) Biochemical

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38. In citrate test, positive result leads to _____ of bromothymol blue to blue. c) Alkalinization

39. Biochemical tests analyze bacterial _____ properties.

c) Metabolic

40. Negative result in methyl red test is indicated by _____ coloration. b) Yellow

41. Colonies greater than 5mm diameter generally indicate _____ bacteria. c) Motile

42. Chemical sterilization methods include _____. a) Ozone

43. Temperature and contact time are critical factors influencing efficacy of _____. a) Moist heat sterilization

44. Early exponential phase marks the beginning of _____ phase in bacterial growth. d) Log

45. Generation time measures rate of _____ in a microbial population. c) Doubling

46. Agar concentration of _____ is used to prepare solid media. c) 1%

47. _____ stains distinguish between gram positive and gram negative bacteria. d) Differential

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48. Anaerobic cultivation methods include _____.

d) Anaerobic jars

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49. Shell vials are used in _____ for cell culture experiments.

c) Inverted microscopes